Customer Churn Prediction and Analysis

1. Project Overview

This project analyses customer subscription data to identify churn patterns, estimate customer lifetime value, and build machine-learning models to predict customer churn.

2. Dataset Overview

• Total customers: 800
• Total features: 11 (excluding CustomerID and Churn)
• Churned customers: 334
• Retained customers: 466

3. Business Metrics

• Churn Rate: 41.75%
• Estimated Monthly Revenue Loss: Rs. 250,382.29
• Average Customer Lifetime Value: Rs. 28,017.59
• Average CLV of Churned Customers: Rs. 27,183.31
• Average CLV of Retained Customers: Rs. 28,615.55

4. Machine Learning Models

Two classification algorithms were trained:

1. Logistic Regression
2. Random Forest Classifier

An 80:20 train-test split was used.

Model Performance

Metric	Logistic Regression	Random Forest
Accuracy	0.6375	0.5938
Precision	0.5763	0.5152
Recall	0.5075	0.5075
F1-Score	0.5397	0.5113

Logistic Regression achieved better accuracy and F1-score on the test set.
It was selected for generating the final customer predictions.

5. Churn Analysis by Contract Type

Churn No Yes
ContractType
Month-to-Month 40.97 59.03
One Year 77.21 22.79
Two Year 80.39 19.61

Month-to-month customers showed the highest churn rate.

6. Churn Analysis by Payment Method

Churn No Yes
PaymentMethod
Card 61.11 38.89
Net Banking 58.79 41.21
UPI 55.92 44.08
Wallet 57.29 42.71

UPI users showed the highest churn rate among the payment methods.

7. Churn Analysis by Subscription Type

Churn No Yes
SubscriptionType
Basic 58.99 41.01
Pro 58.04 41.96
Standard 57.68 42.32

Standard subscription customers showed the highest churn rate, although the differences between subscription types were small.

8. Feature Importance

The Random Forest model was used to rank the features by predictive importance.

Feature Importance
num__TotalChargesINR 0.122876
num__TenureMonths 0.121853
num__MonthlyChargesINR 0.119830
num__Age 0.110418
num__LatePayments 0.080698
num__SupportTickets 0.072810
cat__ContractType_Month-to-Month 0.071972

cat__ContractType_One Year 0.034520

cat__ContractType_Two Year 0.028960

cat__Region_South 0.023446

Feature importance indicates how much the model relied on each feature.
It does not establish causation.

9. Business Recommendations

1. Encourage month-to-month customers to consider annual contracts.
2. Investigate payment difficulties among UPI and Wallet users.
3. Monitor customers with behavioural patterns associated with higher churn.
4. Introduce customer retention campaigns and loyalty benefits.
5. Improve customer support and investigate recurring service issues.

10. Conclusion

The dataset showed an overall churn rate of 41.75%.
Logistic Regression performed better than Random Forest on the reported test-set accuracy and F1-score.

The analysis identifies contract type and payment method as useful areas for further customer-retention investigation.

The predictions and metrics should be interpreted in the context of the dataset and the limitations of the trained models.